How-To Library

Informatica

How to Configure Oracle Single Client Access Name (SCAN) in Informatica 10.1.1

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Abstract

Informatica supports connectivity to an Oracle Real Application Cluster (RAC) for the domain, Model Repository Service, and PowerCenter Repository Service. Informatica services can connect to Oracle RAC configured in Connect Time Connection Failover (CTCF) or Fast Connection Failover (FCF) mode. Effective in version 10.1.1, you can use Oracle Single Client Access Name (SCAN) to connect to Oracle RAC. Use SCAN in Oracle RAC to specify a single name for clients to access any Oracle database running in a cluster. When you use SCAN, the Informatica connection configuration does not need to change if you add or remove nodes or databases in the Oracle RAC. This article describes how to configure Informatica for Oracle RAC with the SCAN mode.

Supported Versions

- Informatica 10.1.1

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Overview

You can use Oracle Single Client Access Name (SCAN) configuration in the connection properties for Informatica services to connect to an Oracle Real Application Cluster (RAC).

When you use SCAN to connect to Oracle RAC, Informatica connection uses a single name to access the Oracle RAC. Internally, Oracle uses SCAN to access any database running in the cluster, independent of the active servers in the cluster. Use SCAN in Oracle RAC to allow load balancing and failover for client connections to the database. The SCAN works as a cluster alias for databases in the cluster.

To connect Informatica and Oracle RAC in SCAN mode, perform the following steps:

- Configure the Informatica domain in SCAN mode.
- Configure the Repository Service to connect to Oracle RAC with SCAN.

Supported Versions

Informatica supports certain configurations for connections between the repository and Oracle RAC.

Informatica 10.1.1 supports the following Oracle RAC versions for connections to Oracle RAC: 11gR2, 12c, and 12cR1.

Note: Informatica does not support Oracle RAC features, such as Run Time Connection Failover mode, also called Transparent Application Failover (TAF).

To identify operating system and other Informatica compatibilities, see the Informatica Platform Availability Matrix (PAM).

Configure the Oracle RAC Instance with Oracle SCAN

When you configure the connection between the Informatica domain or Model Repository Service and Oracle RAC, you can configure the Oracle SCAN in the JDBC URL. When you configure the connection between the PowerCenter repository and Oracle RAC, you can add SCAN in the Oracle TNS configuration file.

Configure the Informatica Domain in SCAN Mode

You can configure the Informatica domain to connect to an Oracle RAC in SCAN mode.

1. Shut down the Informatica domain.
2. Configure SCAN in the JDBC connection details to the Oracle RAC instance with the infasetup UpdateGatewayNode command.

   For example:

   ```
   infasetup.sh UpdateGatewayNode
   -cs "jdbc:informatica:oracle://
   scanHost:scanPort;ServiceName=servicename;LoadBalancing=true;
   MaxPooledStatements=100"
   -du user
   -dp password
   -dt ORACLE
   -dn Domainname
   -nn nodename
   -na nodehost:port
   -rst true
   ```

   Tip: You can use the tnsping command on the Oracle instance to confirm the Oracle service name.
3. Run the infaservice.sh startup command to restart each node that runs the Repository Service.

Configure the PowerCenter Repository Service to Connect to Oracle RAC with SCAN

When you configure the connection between the PowerCenter repository and Oracle RAC, you change the Oracle database host and port with SCAN in the Oracle TNS configuration file for repository database. With SCAN, you do not need to set any failover property in the tnsnames.ora file. The failover properties are implicit on the database side, which are part of the SCAN IP configuration.

To use the SCAN mode to configure the Oracle RAC for cluster changes in the backend, ensure that the clients use the Oracle database version 11gR2 or later.

On the Oracle RAC server, add a TNS entry in the tnsnames.ora file for the repository node.

For example, in the tnsnames.ora file, configure the Oracle RAC instance as follows:

```
RAC12C,RAC12C.company.com=
(DESCRIPTION=
    (ADDRESS= (PROTOCOL=TCP) (HOST=scanHost) (PORT=scanPort))
    (CONNECT_DATA=
                (SERVER=DEDICATED)
                (SERVICE_NAME=RAC12C.company.com)
          )
)
```

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